

01



Connectivity Link Multi-Cluster DNS GLB PoC

Delegated CoreDNS design for application
failover across two OpenShift clusters

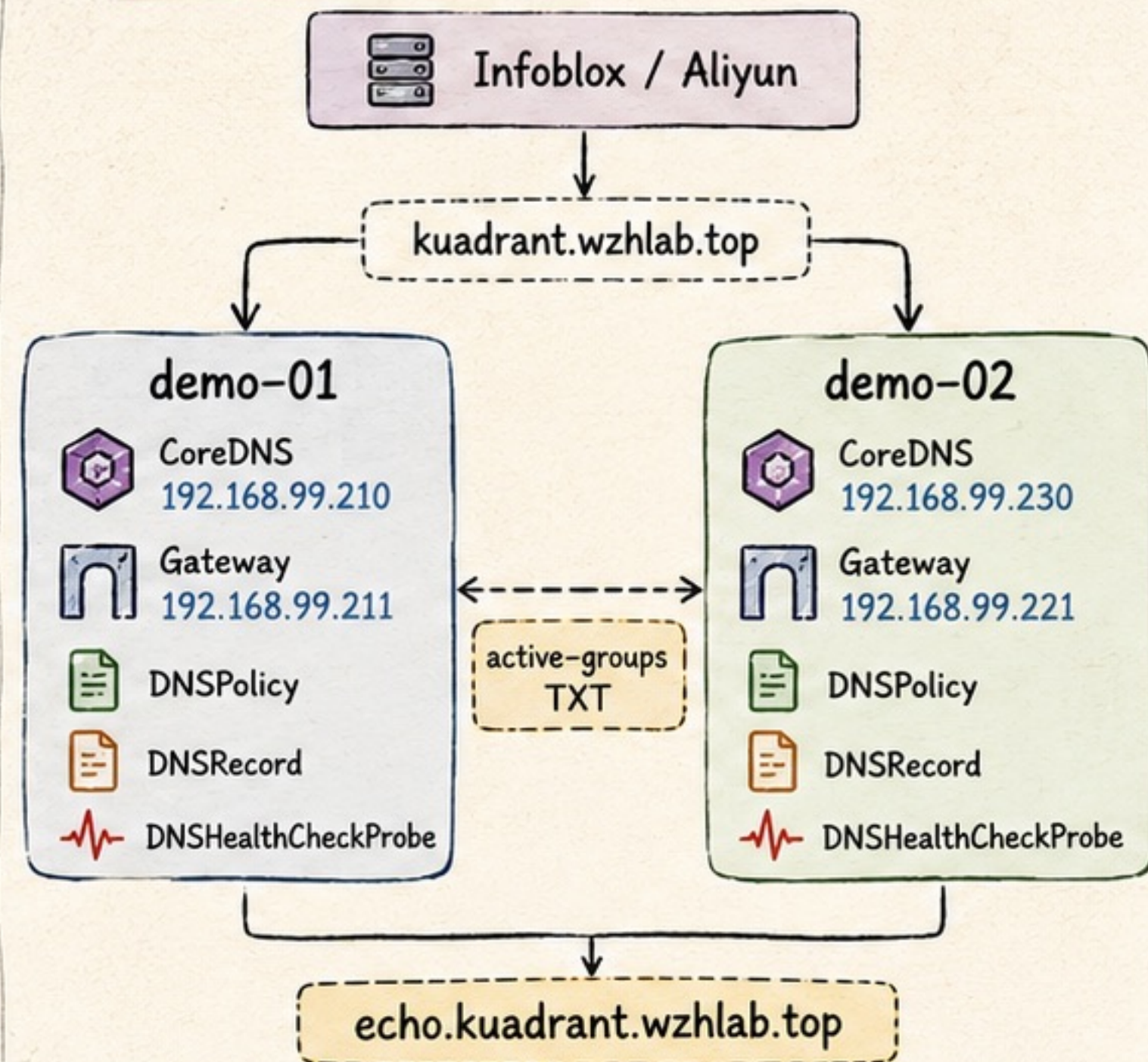


**Validated PoC on real
demo-01 and demo-02 clusters**



Upstream DNS delegates the subdomain.
Cluster CoreDNS answers the application DNS.
Connectivity Link health and DNS policy drive failover.

TOPOLOGY OUTLINE



Author:
George Zheng Wang (SAA)

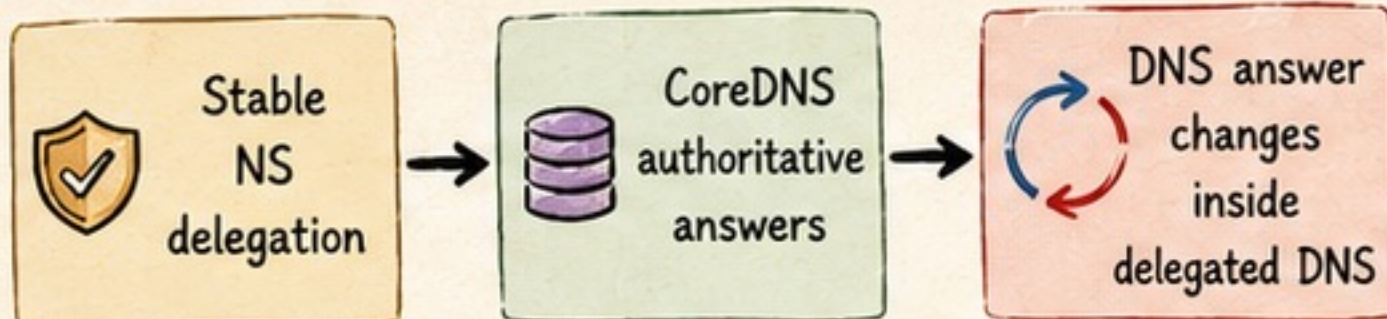


Date:
2026-05-25



Environment:
demo-01 OCP,
demo-02 OCP,
Aliyun DNS simulating
Infoblox delegation

LOGIC FLOW



Connectivity Link DNS GLB PoC



EVIDENCE / IP DETAILS

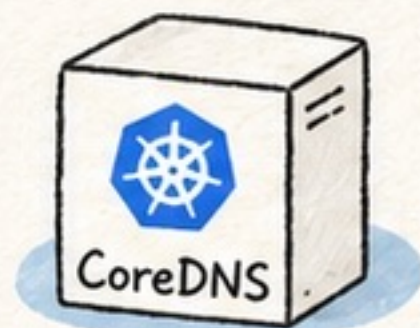
	ns1 / demo-01 CoreDNS:	192.168.99.210
	ns2 / demo-02 CoreDNS:	192.168.99.230
	demo-01 Gateway:	192.168.99.211
	demo-02 Gateway:	192.168.99.221

SCOPE / HIGHLIGHTS



EXECUTIVE SUMMARY

- 1 Use managed-cluster CoreDNS as delegated authoritative DNS endpoints.



- 2 Failover happens by changing RHCL DNS Groups through active-groups TXT, not by changing upstream NS records.

active-groups TXT
"version=1;
groups=demo-01&&demo-02"

- 3 The validated result is that both delegated CoreDNS authorities return only the healthy gateway only during demo-02 failure.



CORE LOGIC

Stable NS delegation
Upstream DNS (Infoblox / Aliyun) delegates kuadrant.wzhlab.top



DNS answer changes inside delegated DNS
Delegated CoreDNS authorities update DNSRecord based on active-groups TXT



Connectivity Link health state drives which gateway IP is returned
DNSHealthCheckProbe feeds the controller

DECISION STRIP

PHASE 1 RECOMMENDATION

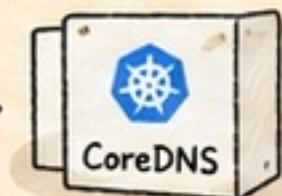
Infoblox / Aliyun



Delegate

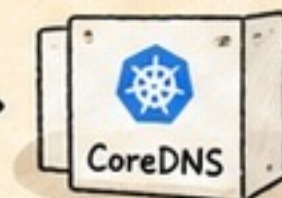
kuadrant.wzhlab.top (subdomain)

ns1.kuadrant.wzhlab.top



192.168.99.210

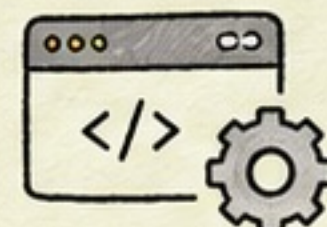
ns2.kuadrant.wzhlab.top



192.168.99.230

Stable NS delegation

FUTURE OPTION



Controller

Write via DTC or WAPI

Infoblox / Aliyun



Enterprise DNS

Controller writes Infoblox DTC or WAPI

EVIDENCE SNAPSHOT

! demo-02 failure:

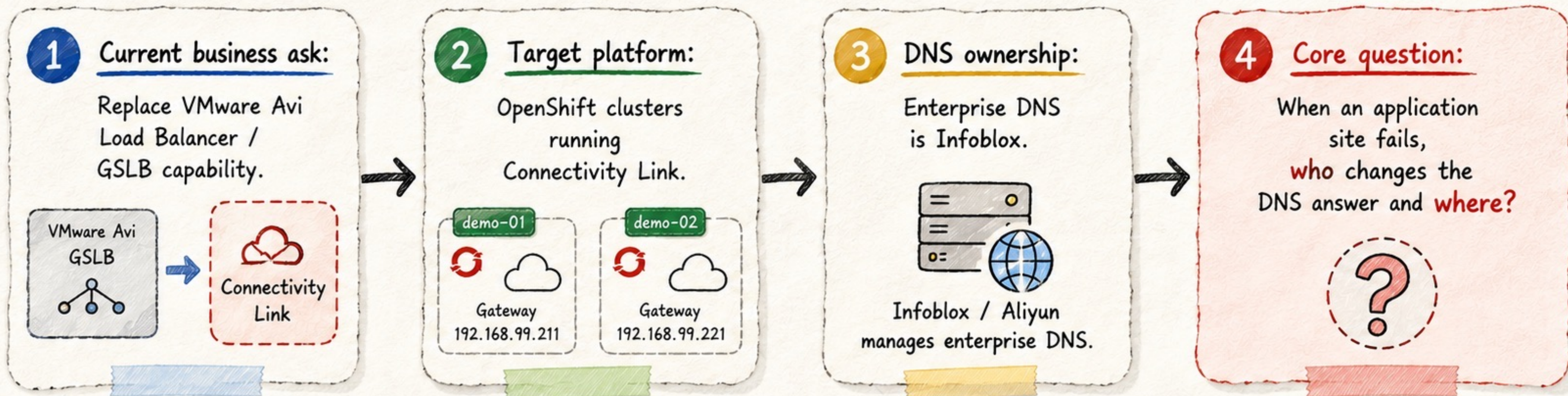
@192.168.99.210 echo.kuadrant.wzhlab.top → 192.168.99.211
@192.168.99.230 echo.kuadrant.wzhlab.top → 192.168.99.211



Both CoreDNS authorities return only the healthy gateway (demo-01) during demo-02 failure.

BUSINESS CONTEXT

Why this PoC matters and the core question we are answering



The PoC focuses on DNS failover behavior, not generic application routing.



Connectivity Link

Gateway / HTTPRoute decide how traffic is routed **after** it arrives.



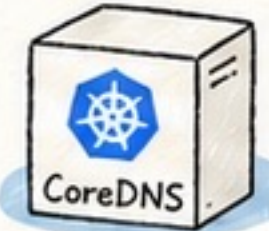
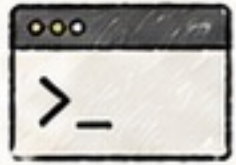
GLB DNS

Decides which entry IP the client receives **first**.

CONSTRAINTS



Our design is driven by delegation **constraints**, not preference.

CONSTRAINT		DESIGN IMPACT	
1	 Infoblox only delegates a subdomain in phase 1 (kuadrant.wzhlab.top)	→	 OCP side must run authoritative DNS for that subdomain.
2	 Controller is not allowed to modify upstream DNS yet	→	 NS records stay stable during failover.
3	 Need cross-cluster application failover	→	 DNS answer must remove unhealthy cluster endpoint.
4	 Need evidence from real clusters	→	 PoC must show command output and DNS response changes.



Stable upstream delegation means failover **cannot rely on** changing parent NS records.

Therefore, the switching logic must happen **inside the delegated authoritative DNS layer.**



EVIDENCE: STABLE NS DELEGATION



kuadrant.wzhlab.top.	NS	ns1.kuadrant.wzhlab.top.
kuadrant.wzhlab.top.	NS	ns2.kuadrant.wzhlab.top.

Stable NS delegation → DNS answer changes **inside delegated DNS.**



FINAL ARCHITECTURE

Delegated CoreDNS design for application failover across two OpenShift clusters

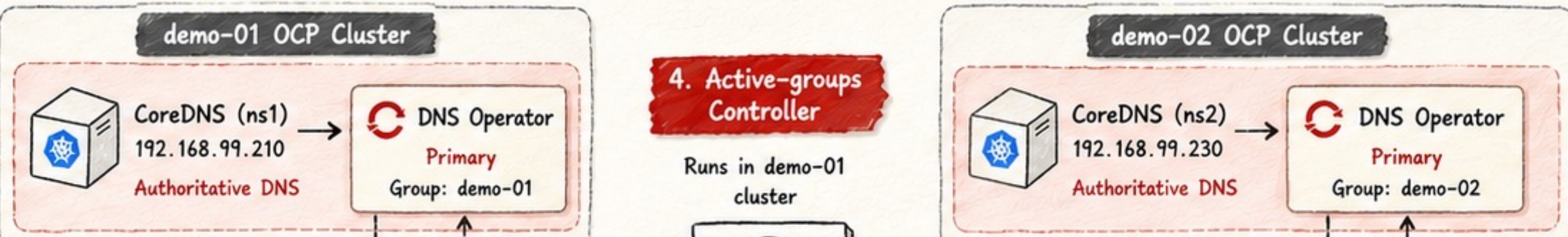
1. Upstream DNS

Stable NS delegation by Infoblox / Aliyun



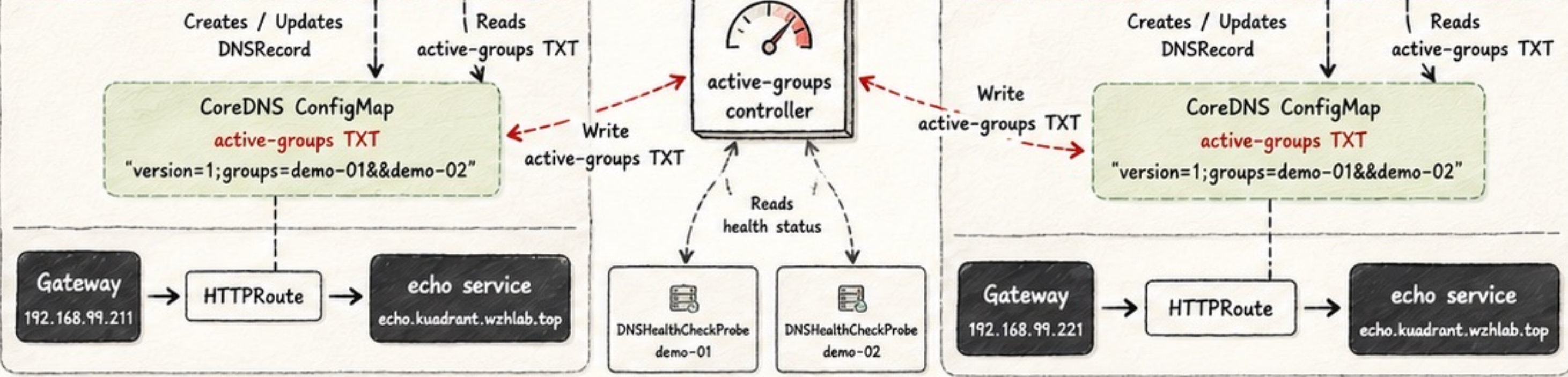
2. Authoritative DNS

CoreDNS in each managed OCP cluster authoritatively answers application DNS.



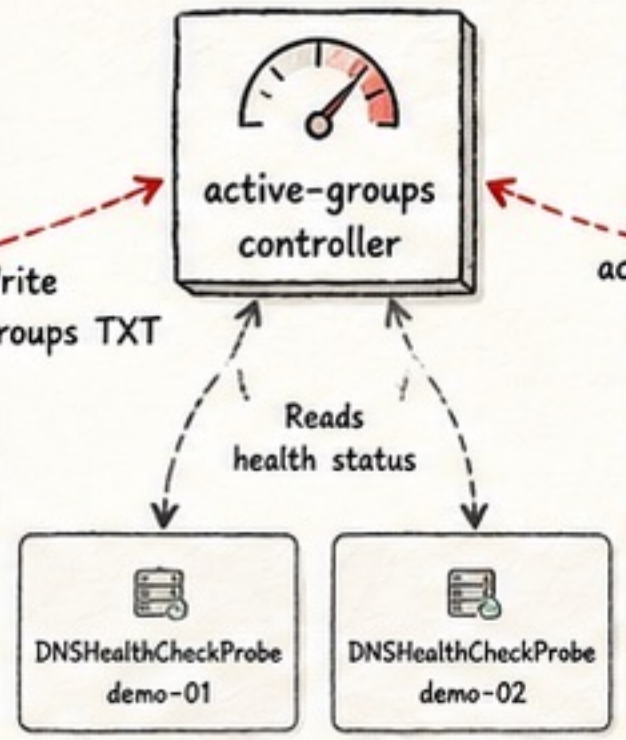
3. Ingress & App

Gateway routes traffic to the echo application via HTTPRoute.



4. Active-groups Controller

Runs in demo-01 cluster



Key IP Addresses

- ns1 CoreDNS: 192.168.99.210
- ns2 CoreDNS: 192.168.99.230
- demo-01 Gateway: 192.168.99.211
- demo-02 Gateway: 192.168.99.221

Key DNS Names

- Delegated subdomain: **kuadrant.wzhlab.top**
- Application host: **echo.kuadrant.wzhlab.top**
- NS records (stable):
ns1.kuadrant.wzhlab.top
ns2.kuadrant.wzhlab.top

Flow Summary

- Client query → upstream DNS (Stable NS delegation)
- Query → CoreDNS authority (ns1 or ns2)
- CoreDNS answers → selected Gateway IP
- Gateway → HTTPRoute → echo service
- Controller updates active-groups TXT → DNS Operator recalculates DNSRecord



MAIN IDEA: Upstream NS delegation stays stable. → DNS answer changes inside delegated CoreDNS. → Connectivity Link health state (via DNSHealthCheckProbe) drives which Gateway IP is returned.

WHY NS POINTS TO MANAGED CLUSTER COREDNS

Clear separation of DNS authority from application entry

1 PARENT ZONE



Infoblox / Aliyun
parent zone

- ✓ Infoblox or Aliyun parent zone
- ✓ Delegates kuadrant.wzhlab.top
- ✓ Keeps NS records stable during failover

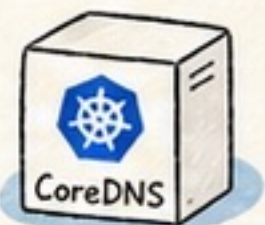
NS delegation
(stable)

kuadrant.wzhlab.top. NS
ns1.kuadrant.wzhlab.top.
kuadrant.wzhlab.top. NS
ns2.kuadrant.wzhlab.top.

2 DELEGATED AUTHORITATIVE DNS

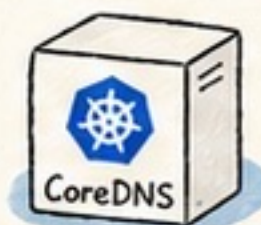
demo-01 CoreDNS

192.168.99.210



demo-02 CoreDNS

192.168.99.230



Serves kuadrant.wzhlab.top
authoritative answers

DNS answer
(with Gateway IP)

A/AAAA for
echo.kuadrant.wzhlab.top
192.168.99.211
192.168.99.221
(or only the healthy one)

3 APPLICATION ENTRY

demo-01 Gateway

192.168.99.211



demo-02 Gateway

192.168.99.221



Returned or removed based on
active group and health

CLIENT



Query:
echo.kuadrant.wzhlab.top

1

Recursive DNS
query

PARENT DNS
(Infoblox / Aliyun)



Holds NS records only
for kuadrant.wzhlab.top

2

Query forwarded to
one of the NS targets

demo-01 CoreDNS

192.168.99.210



demo-02 CoreDNS

192.168.99.230



Authoritative answer for
echo.kuadrant.wzhlab.top

3

Returns selected
Gateway IP(s)

GATEWAY

demo-01 Gateway
192.168.99.211



demo-02 Gateway
192.168.99.221



HTTPRoute
→ echo service

APPLICATION



echo service
(echo.kuadrant.wzhlab.top)



NS targets are
DNS authorities.

Gateway IPs are
application traffic targets.

The failover decision changes
DNS answers, not the parent delegation.

ENVIRONMENT EVIDENCE

Validated on two freshly installed OpenShift clusters



Environment
PoC Lab
Clean & Newly Deployed



Clusters
2
demo-01, demo-02



CoreDNS Authorities
2
Managed in OCP



Gateways
2
Application Entry Points



Overall Health
Healthy
All systems operational

Environment Overview

Area	demo-01	demo-02
 Nodes	192.168.99.23/24/25 Ready 	192.168.99.33/34/25 Ready 
 Client	oc 4.20.21 	oc 4.20.21 
 User	system:admin 	admin 
 ClusterOperators	all Available 	all Available 

demo-01

```
$ oc get clusteroperators
NAME                                STATUS    AVAILABLE   AGE
authentication                      Available True        2d10h
cloud-controller-manager            Available True        2d10h
etcd                                Available True        2d10h
...
13 clusteroperators
```



demo-02

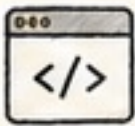
```
$ oc get clusteroperators
NAME                                STATUS    AVAILABLE   AGE
authentication                      Available True        2d10h
cloud-controller-manager            Available True        2d10h
etcd                                Available True        2d10h
...
13 clusteroperators
```



Takeaway: Two clean and healthy OpenShift clusters.
This PoC data is generated from this round of deployment.



Delegated subdomain:
kuadrant.wzhlab.top



Application host:
echo.kuadrant.wzhlab.top

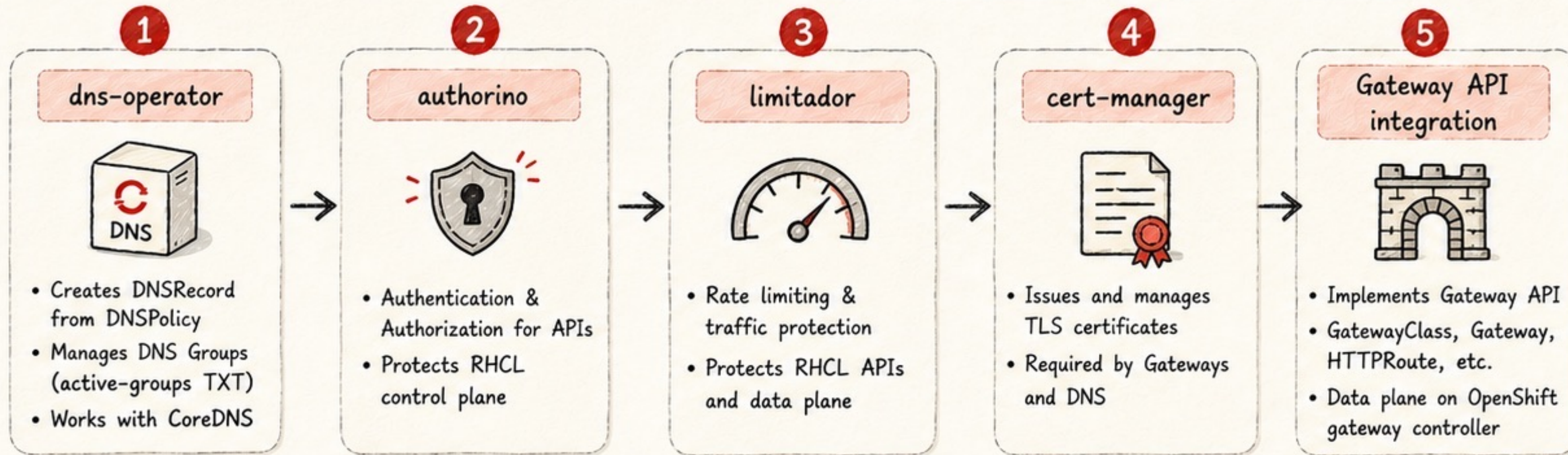


Upstream DNS provider (simulated):
Infoblox / Aliyun

RHCL INSTALLATION



RHCL is composed of multiple Operators.
Each has a clear role in the control plane.



These components work together under **Kuadrant** to deliver application entry, policy, and DNS capabilities.

Installation Evidence (demo-01)

```

$ oc get pods -n kuadrant-system
NAME                                READY   STATUS    RESTARTS   AGE
dns-operator-7f6b7b7c9d-kwt8q       1/1     Running   0           2d10h
authorino-6c676d8d5f-r8q2m          1/1     Running   0           2d10h
limitador-7c7589f9c9-2g6k5          1/1     Running   0           2d10h
cert-manager-7b46c6f7d7-m8sx2       1/1     Running   0           2d10h
controller-manager-6b7f7d5f4c-n9t2  1/1     Running   0           2d10h
gateway-api-controllers-6fdd4c8f7f-gkq7z 1/1     Running   0           2d10h
  
```

```

$ oc get kuadrant -n kuadrant-system kuadrant -o wide
NAME      VERSION   READY   AGE
kuadrant  1.2.0     True    2d10h
  
```

Kuadrant Ready=True
All Operators are healthy

Core Resources

```

kind: GatewayClass
metadata:
  name: openshift-default
spec:
  controllerName: openshift.io/
    gateway-controller/v1
  
```

```

kind: Kuadrant
metadata:
  name: kuadrant
spec:
  # defaults
  
```



GatewayClass defines the implementation of Gateway API on OpenShift.



Kuadrant activates the RHCL control plane and all installed Operators.



Clean environment with all core RHCL components healthy. → Ready to create **DNSPolicy**, **DNSRecord**, and **DNSHealthCheckProbe** for multi-cluster DNS GLB.

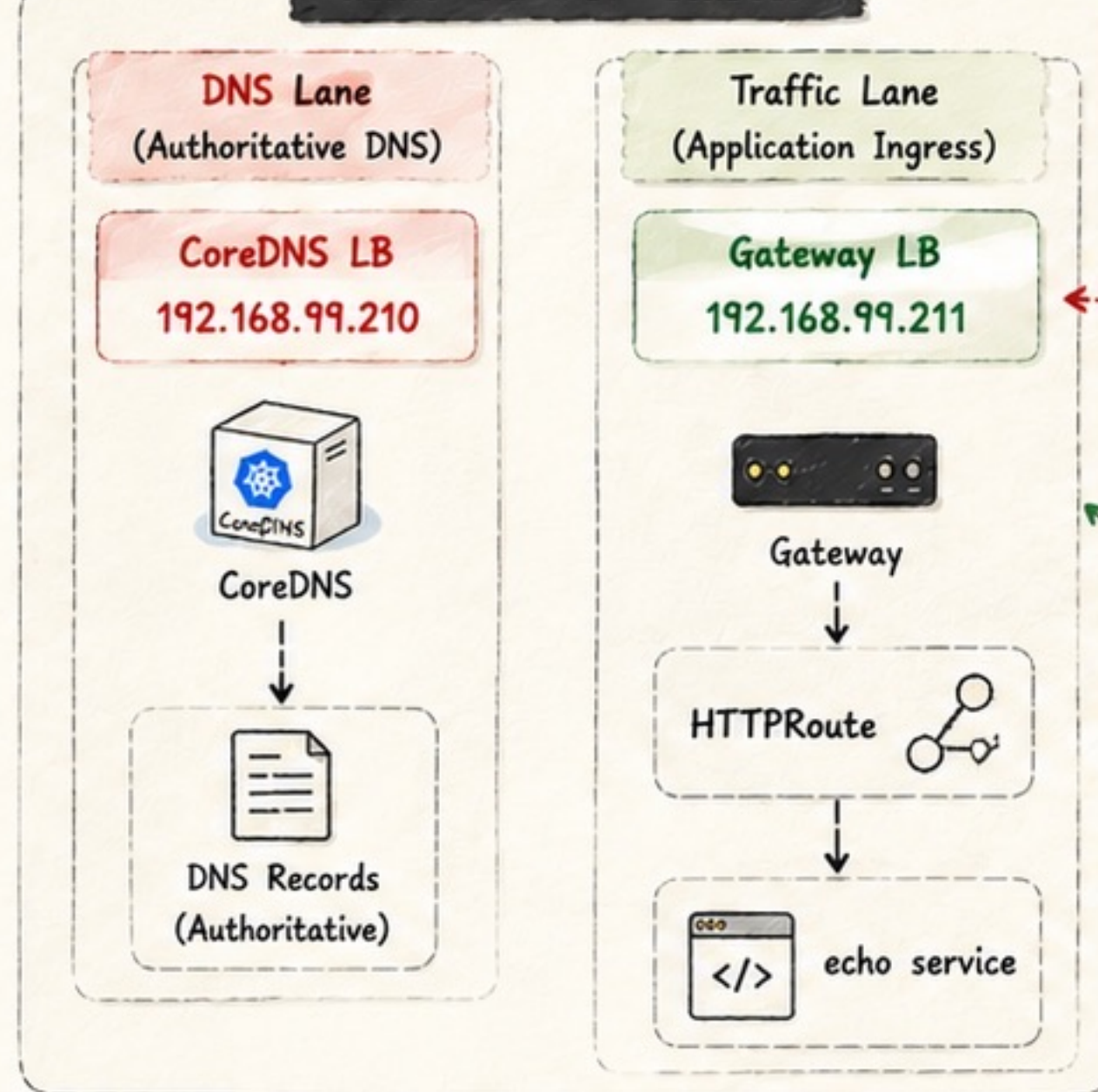
LOADBALANCER IP PLAN

Two IPs per cluster for different purposes – keep DNS and traffic separate

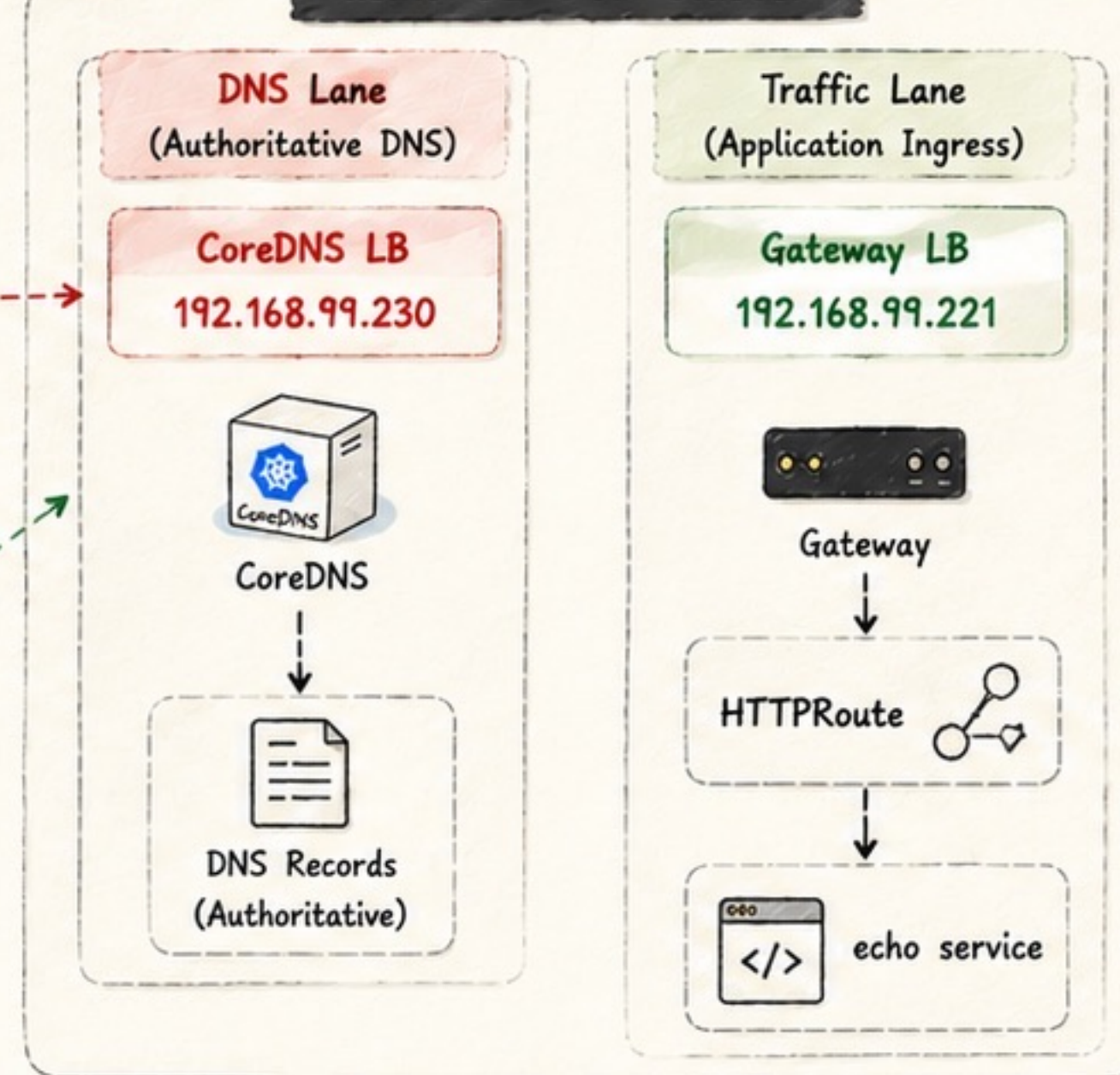
End-to-end Flow

- Client resolves `echo.kuadrant.wzhlab.top`
- Upstream DNS (Infoblox / Aliyun) delegates `kuadrant.wzhlab.top` (Stable NS delegation)
- Authoritative DNS (CoreDNS in OCP) returns a Gateway IP (based on health)
- Client connects to Gateway IP via HTTP
- Traffic routed to echo application

demo-01 OCP Cluster



demo-02 OCP Cluster



IP Plan Summary

Role	demo-01	demo-02	Purpose
CoreDNS LB	192.168.99.210	192.168.99.230	NS glue target / authoritative DNS
Gateway LB	192.168.99.211	192.168.99.221	Application ingress (HTTP traffic)

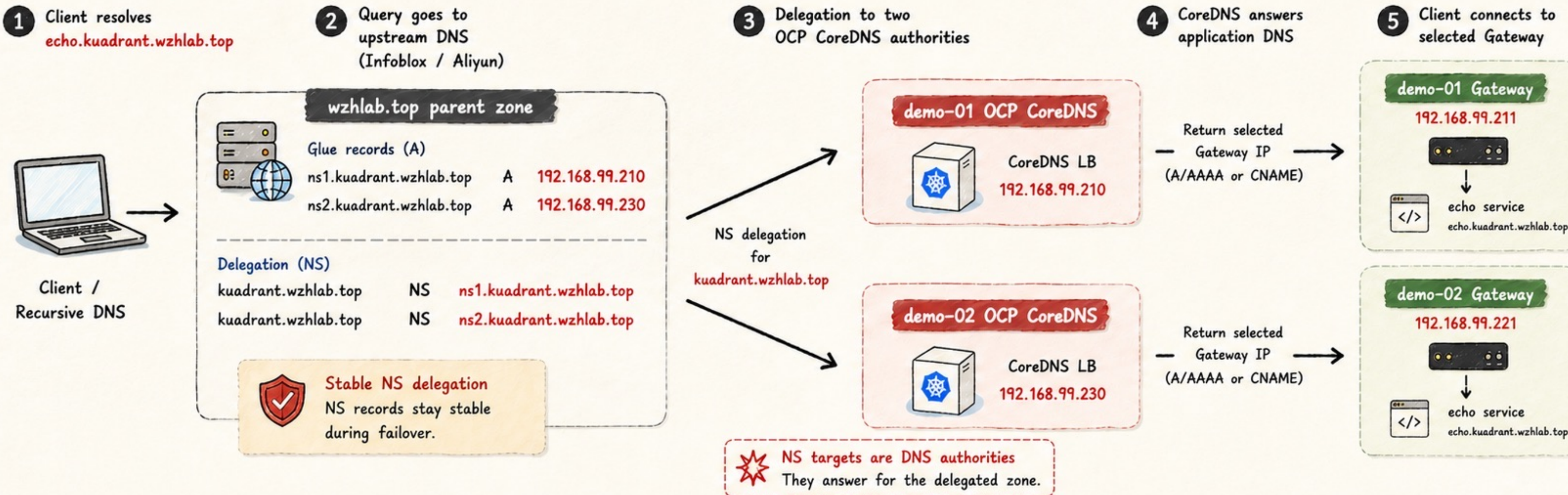


Key Takeaway

- DNS Lane IPs (CoreDNS) are for **authoritative DNS** only.
- Traffic Lane IPs (Gateway) are for application **HTTP traffic**.
- Health controls which **Gateway IP** is returned by DNS.

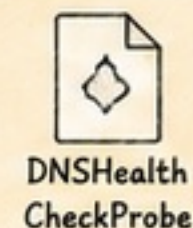
Simulating Infoblox Delegation with Aliyun DNS

Parent DNS only delegates the subdomain. Failover happens inside OCP CoreDNS, not in the parent zone.



Parent zone (Infoblox / Aliyun) only delegates:
`kuadrant.wzhlab.top` → ns1/ns2 on OCP CoreDNS

Application DNS records (e.g., `echo.kuadrant.wzhlab.top`)
are answered inside CoreDNS based on health and active-groups TXT.



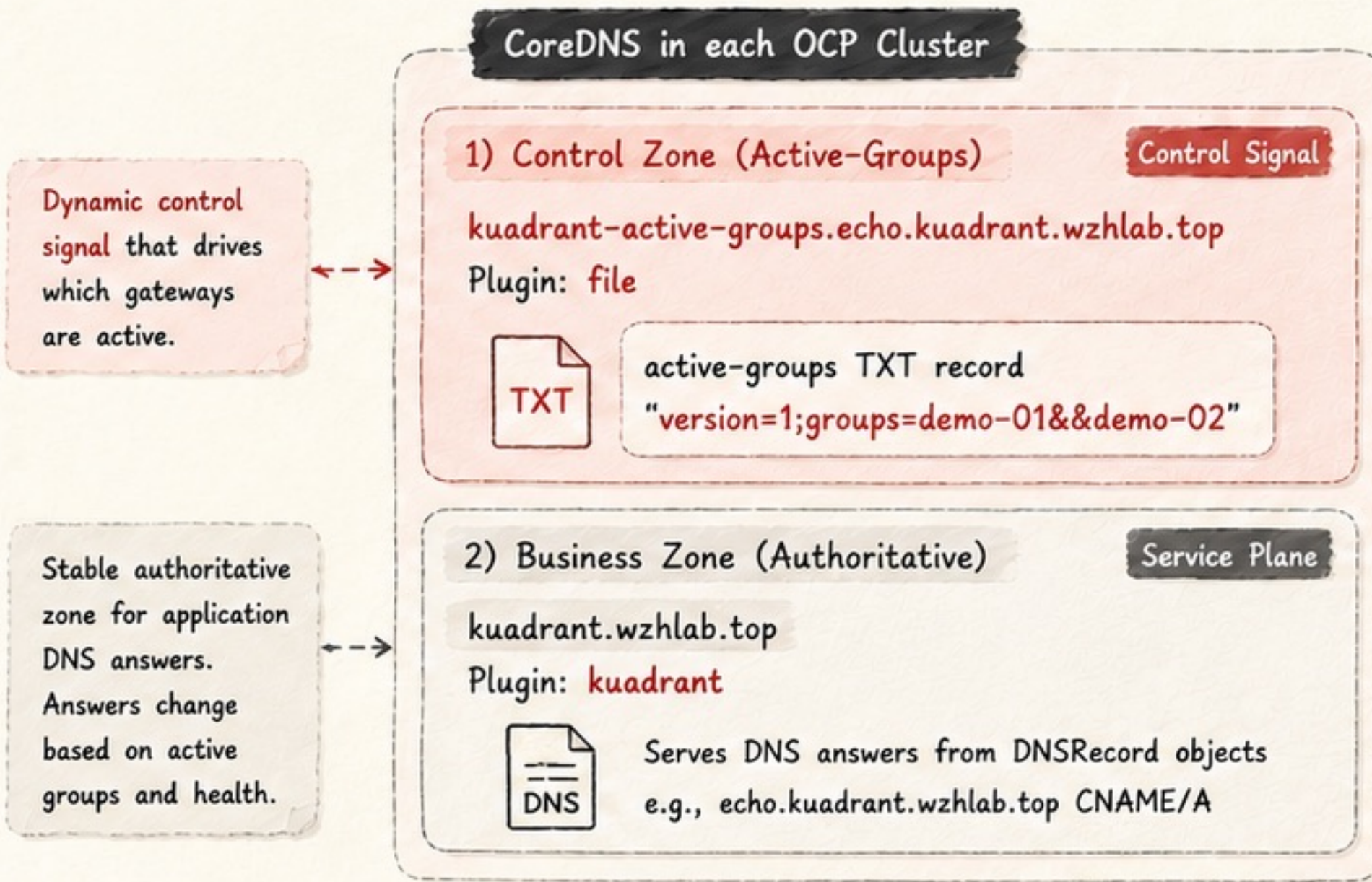
In this PoC, Aliyun DNS simulates Infoblox delegation behavior.



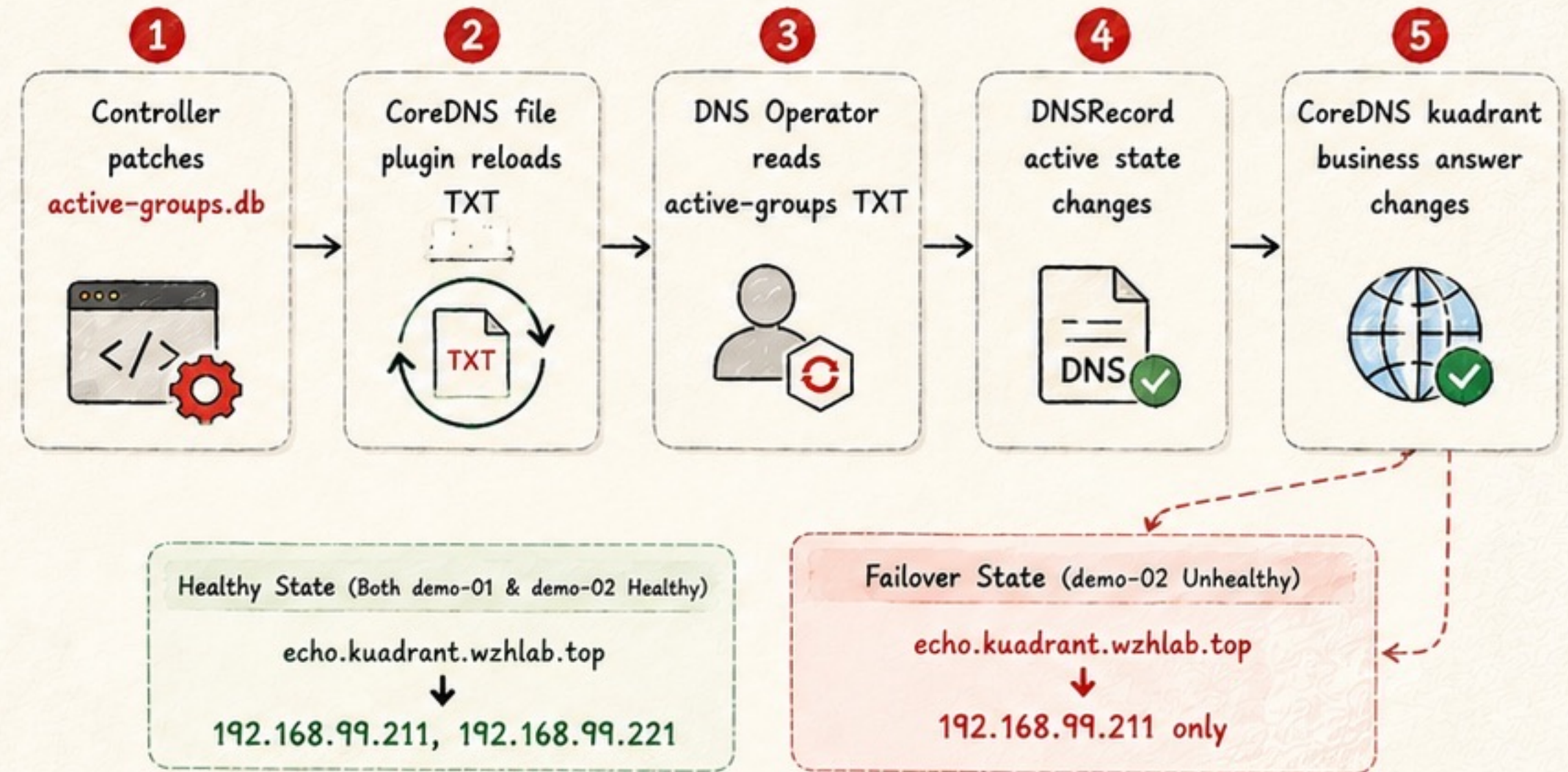
Key point: The parent DNS **does not change** during failover.
DNS answer changes happen **inside the delegated CoreDNS authorities**.

CoreDNS Design

Two zones, two responsibilities – control signal and business service separation



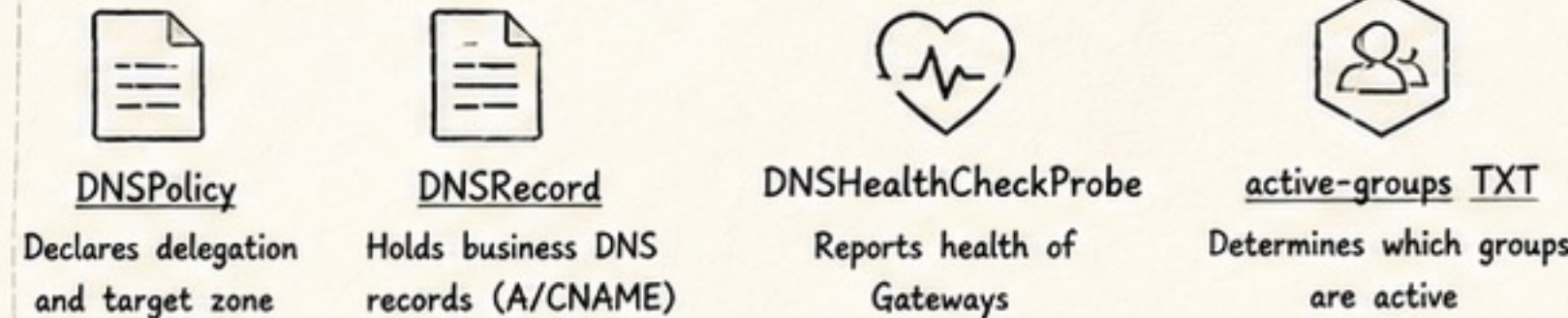
Internal Control Chain (How Switching Happens)



Key Point

The **active-groups TXT** is the control signal. Business DNS answers are generated from **DNSRecord** objects.

Related RHCL Resources



Example CoreDNS ConfigMap

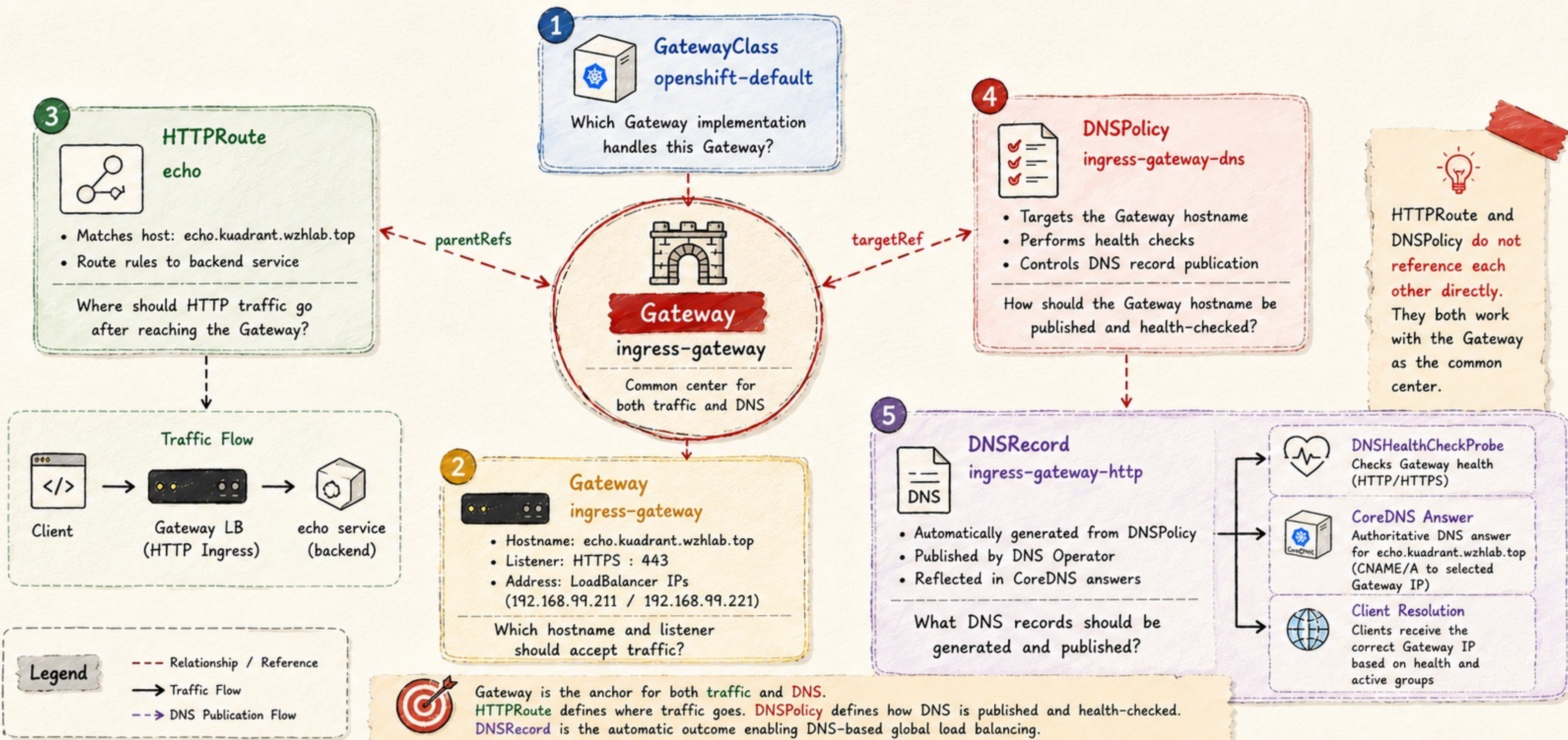
Control Zone (file plugin): kuadrant-active-groups.echo.kuadrant.wzhlab.top	}	Serves TXT control signal
Business Zone (kuadrant plugin): kuadrant.wzhlab.top		
	}	Serves business DNS answers



Stable NS delegation from Infoblox/Allyun points to CoreDNS IPs. Failover is achieved by **changing DNS answers inside CoreDNS**, not by changing parent NS.

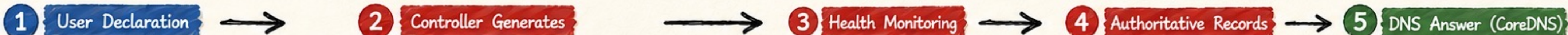
Gateway, HTTPRoute, DNSPolicy

Objects work around the Gateway. Traffic routing and DNS publishing are decoupled.



DNSPolicy to DNSRecord

From user intent to global DNS answers – RHCL generates and manages everything.



DNSPolicy (User Created)

```
kind: DNSPolicy
metadata:
  name: ingress-gateway-dns
spec:
  delegate: true
  healthCheck:
    protocol: HTTP
    port: 80
    path: /health
    interval: 30s
    failureThreshold: 2
  targetRef:
    name: ingress-gateway
    kind: Gateway
  domains:
    - echo.kuadrant.wzhlab.top
```

DNSRecord (Controller Generated & Managed)

Name: ingress-gateway-http

Spec

rootHost:
echo.kuadrant.wzhlab.top

endpoints (chain):

- 1 CNAME echo.kuadrant.wzhlab.top → klb.echo.kuadrant.wzhlab.top
- 2 CNAME klb.echo.kuadrant.wzhlab.top → geo-na.klb.echo.kuadrant.wzhlab.top
- 3 A geo-na.<hash>.klb.echo.kuadrant.wzhlab.top → <Gateway IP>

Status (Managed by RHCL)

- activeGroups:
 - demo-01
 - demo-02
- group: geo-na
- ownerID: 4f8c2a1e-0e6d-4c49-8b1a-2c9d5e3f7b11
- state: **Healthy**
- lastTransitionTime: 2025-05-16T12:30:10Z

RHCL controller creates and updates DNSRecord based on DNSPolicy, active-groups signal, and health.

DNSHealthCheckProbe (Generated)



- Protocol: HTTP
- Port: 80
- Path: /health
- Interval: 30s
- FailureThreshold: 2
- Targets:
 - demo-01 Gateway 192.168.99.211
 - demo-02 Gateway 192.168.99.221

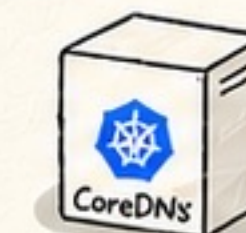
Result: **Healthy**
(Both gateways healthy)

authoritative-record (Generated)



Records pushed to CoreDNS provider (kuadrant plugin):

- 1 CNAME
echo.kuadrant.wzhlab.top → klb.echo.kuadrant.wzhlab.top
- 2 CNAME
klb.echo.kuadrant.wzhlab.top → geo-na.klb.echo.kuadrant.wzhlab.top
- 3 A
geo-na.<hash>.klb.echo.kuadrant.wzhlab.top → 192.168.99.211 192.168.99.221 (active both)



Client query:
echo.kuadrant.wzhlab.top

Answer returned:

○ CNAME
→ klb.echo.kuadrant.wzhlab.top

○ CNAME
→ geo-na.klb.echo.kuadrant.wzhlab.top

○ A
→ 192.168.99.211 192.168.99.221 (active both)

User intent

What should be published and how to check health.



Generated objects

RHCL produces a chain of records and maintains status and ownership.



Health signal

Probes validate gateways from outside the cluster.



Publish

Records are written to the authoritative DNS data.



DNS answer

CoreDNS returns the current active Gateway IPs.



RHCL does not create a single A record manually. It builds a multi-layer CNAME/A chain, maintains health, group and ownership, and publishes the correct answers automatically.

Why Two DNSRecords Appear in Console

Not duplicate configuration — two layers of delegated DNS.

1 Business DNSRecord

`ingress-gateway-http`

 Role	Manages business DNS for <code>echo.kuadrant.wzhlab.top</code>
 Health field	<code>Healthy=True/False</code>
 Key status fields	• activeGroups: <code>demo-01</code>, <code>demo-02</code> • group: <code>geo-na</code> • ownerID: <code>4f8c2a1e-0e6d-4c49-8b1a-2c9d5e3f7b11</code>
 Troubleshooting use	Check application DNS health, activeGroups, group

VS

2 Authoritative Publication Record

`authoritative-record-zzy9f4tx`

 Role	Publishes final DNS set to CoreDNS (authoritative)
 Health field	<code>Healthy = -</code> (Not an application health object)
 What it contains	Final CNAME / A / TXT records that are written to CoreDNS
 Troubleshooting use	Verify the exact DNS records (CNAME / A / TXT) published to CoreDNS

Controller Relationship

Checks by RHCL



- HTTP health probes
- Gateway reachability
- Active groups from active-groups TXT
- Endpoint (Gateway) health

Drives `Healthy-True/False` and active state

Controller creates and maintains the authoritative-record based on the `ingress-gateway-http` and `DNSPolicy`.

Publishes to CoreDNS



CoreDNis

- CNAME chain
- A records (per active group)
- TXT records (if applicable)

Used by clients when resolving DNS answers



Two records are expected in delegated CoreDNS mode.



Key Takeaways

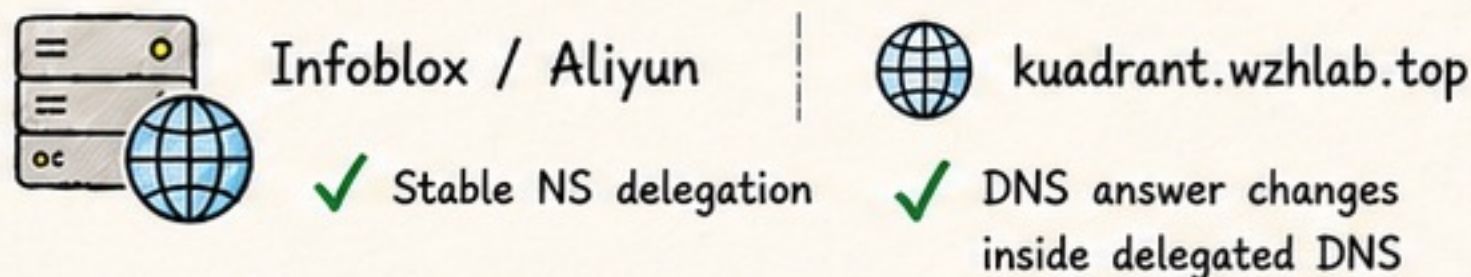
- ✓ Business health? Check `ingress-gateway-http`.
- ✓ Wrong DNS data published? Check `authoritative-record-zzy9f4tx`.
- ✓ `Healthy = -` is normal for `authoritative-record`.



Different purpose, different scope: `ingress-gateway-http` (business health & state) + `authoritative-record-zzy9f4tx` (final DNS publication). Together they deliver reliable **DNS Global Load Balancing**.

CNAME Chain and TXT Ownership

Why DNS records look layered instead of a single A record



CNAME chain enables:

- multi-cluster aggregation
- Geo or weighted routing
- DNS Groups
- ownership isolation



Controller Inputs (Drive the chain & final answer)



TXT ownership (provider-side)

```
kuadrant-<hash->cname-echo.kuadrant.wzhlab.top
heritage=external-dns
external-dns/group=demo-01
external-dns/owner=2omrld2x
external-dns/version=1
```



Not for business clients



Prevents multiple controllers or clusters from overwriting each other

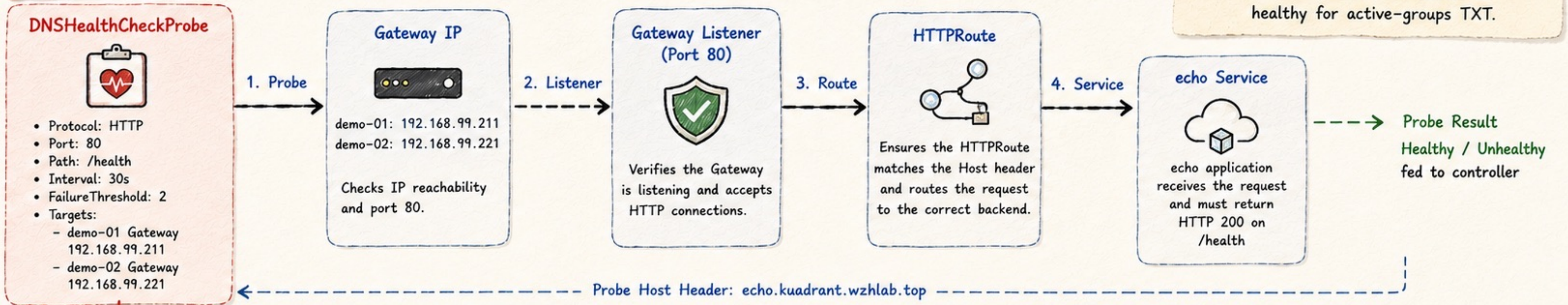


Manager takeaway: this is not duplicate config. The CNAME chain organizes routing layers, and TXT ownership protects record ownership.

DNSHealthCheckProbe

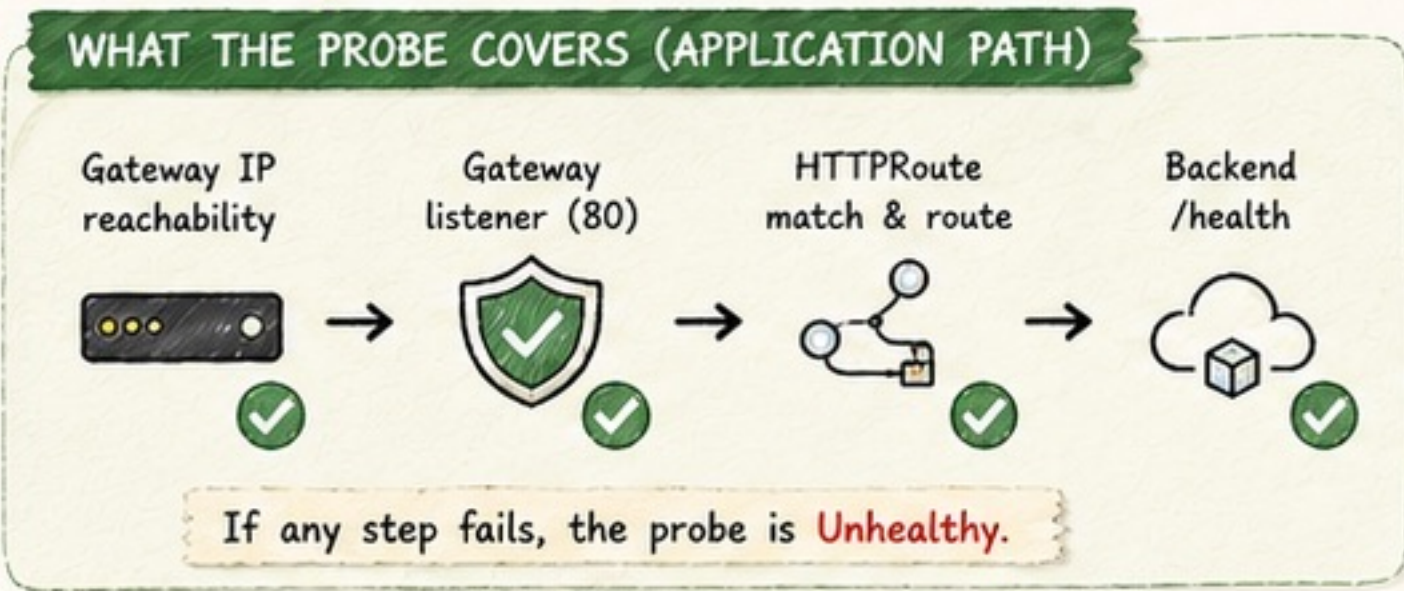
The probe validates the application path behind the Gateway, not DNS infrastructure.

Probe Target Path



Key idea

DNSHealthCheckProbe checks the HTTP health of **echo.kuadrant.wzhlab.top** through the Gateway. It determines which cluster is healthy for active-groups TXT.



HOW TO VALIDATE THOSE AREAS

CoreDNS answers?	<code>dig @<coredns-ip> echo.kuadrant.wzhlab.top</code>
DNSRecord status?	Check DNSRecord (A/CNAME/TXT) status: Healthy / Synced
CoreDNS logs?	<code>kubectl logs -n kuadrant-system deploy/kuadrant-coredns</code>

In short:

DNSHealthCheckProbe validates application path health only. It does not validate DNS infrastructure or delegation.

Demo Environment

CoreDNS (NS targets)		Gateways (probe targets)	
ns1.kuadrant.wzhlab.top	192.168.99.210	demo-01 Gateway	192.168.99.211
ns2.kuadrant.wzhlab.top	192.168.99.230	demo-02 Gateway	192.168.99.221

★ Stable NS delegation
DNS answer changes **inside** delegated DNS

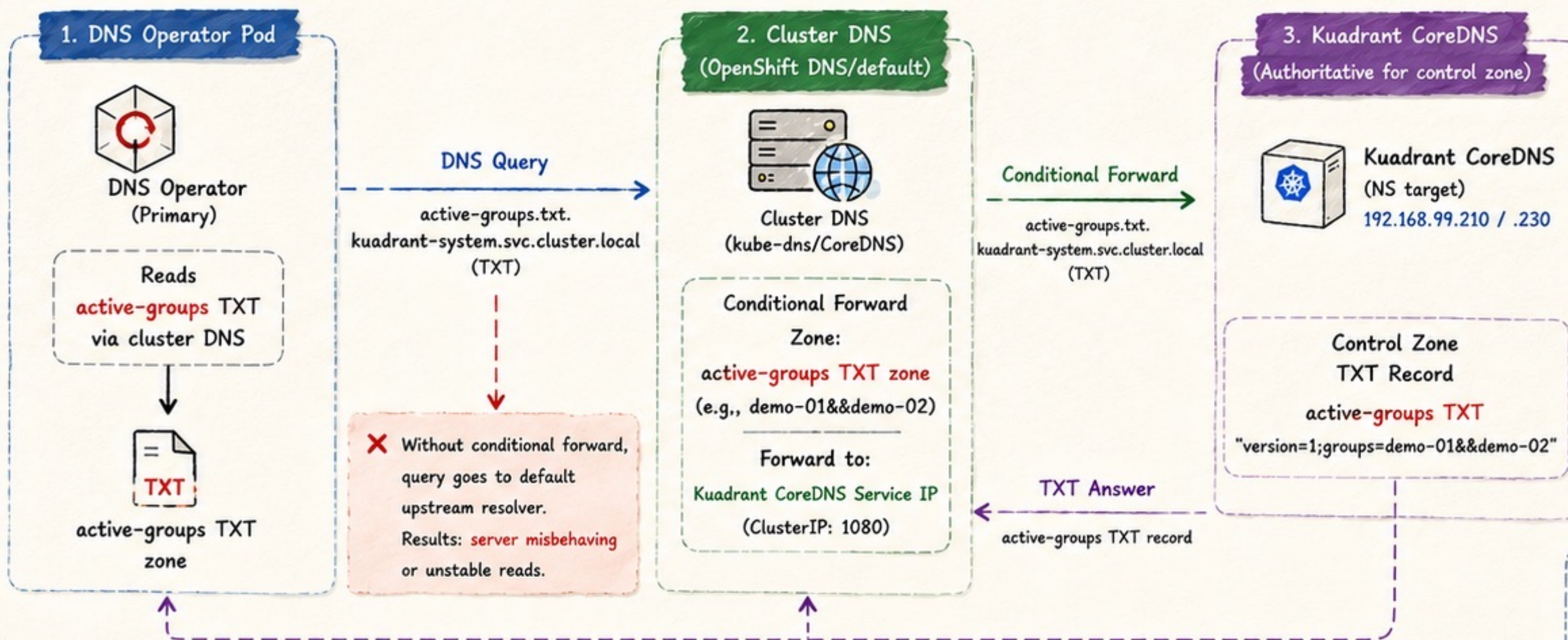
DNS Operator Active Groups



This is a narrow conditional forward for the **active-groups control zone**, not a public DNS change for the application hostname.



Scope:
active-groups zone only



What this configuration does

- ✓ Lets the DNS Operator reliably read the active-groups TXT from local Kuadrant CoreDNS.
- ✓ Uses only for the control zone (active-groups TXT), not for the application hostname.

What this configuration does NOT do

- ✗ Does not change public resolution for echo.kuadrant.wzhlab.top.
- ✗ Does not impact other DNS zones in the cluster.



How to verify

- `@<cluster-dns-ip>`
- active-groups.txt.kuadrant-system.svc.cluster.local TXT
- `kubectl logs -n kuadrant-system deploy/kuadrant-dns-operator`

Key Takeaway

This is a targeted, least-privilege DNS change limited to the active-groups control zone. It **enables reliable active group reads and safe DNS failover**.

Control Zone Name (TXT)

active-groups.txt.kuadrant-system.svc.cluster.local
(demo-01 && demo-02)

Authoritative NS Targets (Kuadrant CoreDNS)

ns1.kuadrant.wzhlab.top	192.168.99.210
ns2.kuadrant.wzhlab.top	192.168.99.230



Multi-Cluster Secrets

Two types of secrets, two purposes, two consumers.

1) RHCL Internal Lane (DNS Operator)



Secret: `kuadrant-system/
demo-01 and demo-02`



Used by:
RHCL DNS Operator



Purpose:
Internal multi-cluster DNS delegation
and DNSRecord aggregation.



Multi-Cluster Secrets

`kubectl-kuadrant_dns
add-cluster-secret`

Generates kubeconfig Secret
without requiring any
manual password.

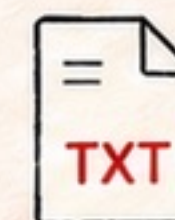
2) Controller Lane (PoC active-groups controller)



Secret: `rhcl-active-groups-
controller/demo-02-kubeconfig`



Used by:
PoC active-groups controller (demo-01)



Purpose:
Read remote DNSHealthCheckProbe
and update active-groups TXT
in both CoreDNS.

How the secrets connect

What does this secret contain?



- Remote API server endpoint
 - Cluster CA data
 - ServiceAccount bearer token
- Used by **RHCL DNS Operator**
to read remote DNSRecord status.



This is not for
any custom
controller.

demo-01 Cluster



RHCL DNS Operator
(Primary: demo-01)

demo-02 Cluster



RHCL DNS Operator
(Primary: demo-02)

Reads remote
DNSRecord status
via Secret
(demo-02)

Reads Probe
status & writes
active-groups TXT
via Secret
(demo-02)

PoC active-groups
controller
(runs in demo-01)

What this secret is used for



- Read DNSHealthCheckProbe status
from remote cluster
- Write active-groups TXT
to Kuadrant CoreDNS



Role is different
from RHCL DNS
Operator.

Command Example

```
$ kubectl-kuadrant_dns add-cluster-secret \
  --from-context demo-02 --to-namespace kuadrant-system
```



Creates kubeconfig Secret in **kuadrant-system**
so **RHCL DNS Operator** can talk to **demo-02**.



Key Takeaway

Two secrets, two responsibilities.
Clear separation avoids permission overreach
and misunderstanding.

Active-Groups Controller



The controller maintains the active-groups TXT control signal.
It **does not** change application DNS records or parent NS delegation.

1 READ (Health Signals)

Reads DNSHealthCheckProbe.status.healthy from each cluster.



demo-01

DNSHealthCheckProbe
status.healthy = True / False



demo-02

DNSHealthCheckProbe
status.healthy = True / False



Why probes, not DNSRecord.Healthy?

DNSRecord.Healthy is affected by active-group membership and can cause recovery lock-in.
Probes reflect real application health.

2 COMPUTE (Decision)

Decides which clusters are active.
Builds the active-groups set.

Example results:

Both healthy:

demo-01&&demo-02

One unhealthy (demo-02 down):

demo-01

Both unhealthy:

(empty)

3 WRITE (Control Signal)

Writes the control signal to both
Kuardant CoreDNS ConfigMaps.

active-groups.db (TXT)

"version=1;groups=demo-01&&demo-02"
(In both clusters)

demo-01 CoreDNS
(NS target)



192.168.99.210

demo-02 CoreDNS
(NS target)



192.168.99.230



WHAT IT DOES

- Reads probe health
- Computes active set
- Writes active-groups TXT
- Triggers reload and reconciliation



WHAT IT DOES NOT DO

- Does not modify application A/CNAME records
- Does not change parent NS delegation
- Does not touch Infoblox / Aliyun upstream DNS

4 TRIGGER (Propagation Flow)



CoreDNS reloads
ConfigMap

Rereads active-groups.db



DNS Operator
reconciles

Recalculates DNSRecord
based on active groups



Authoritative
DNS answers
update

Only healthy cluster
gateway IP(s) returned



Clients receive
updated answers

Traffic goes to
healthy site(s)

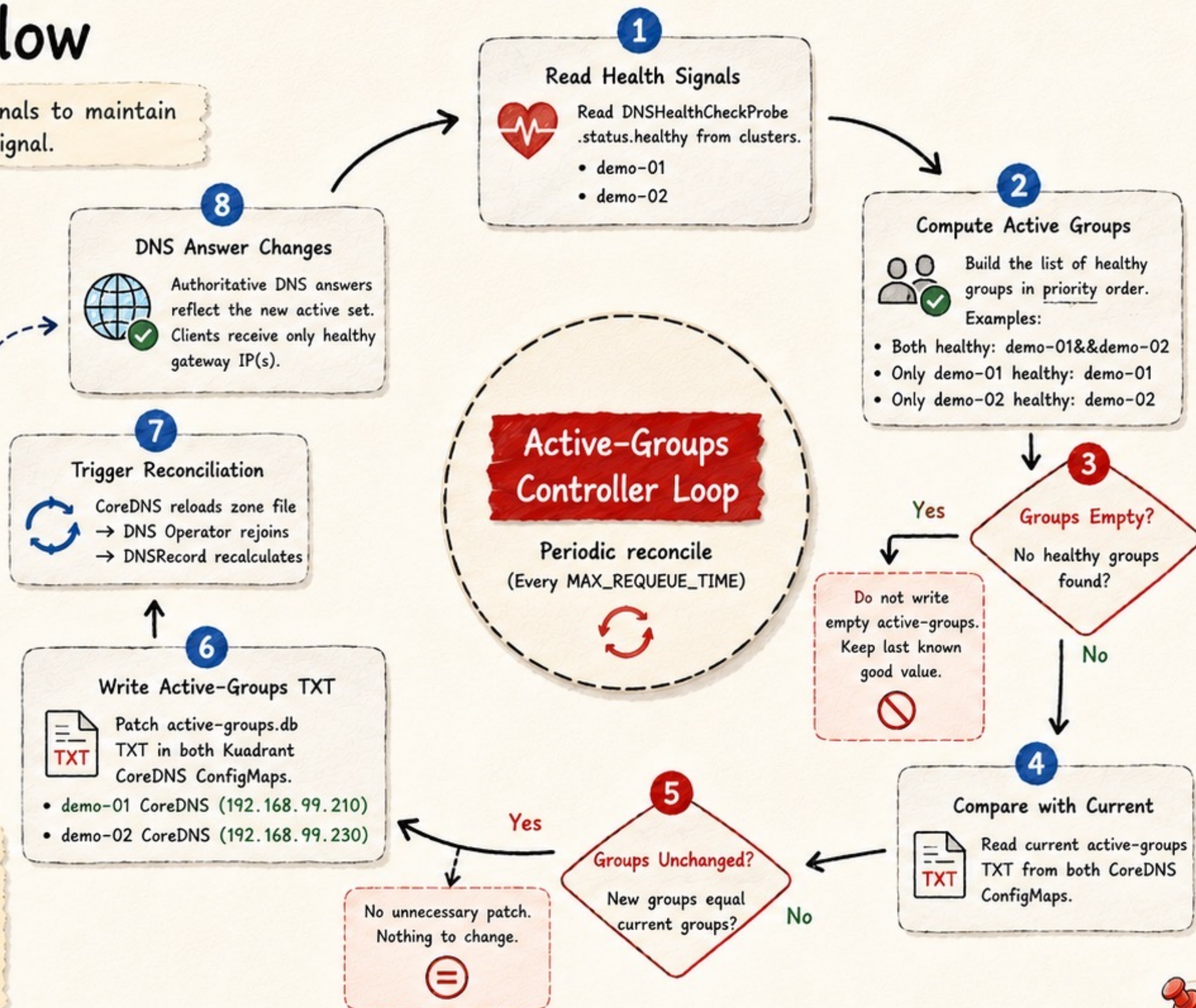
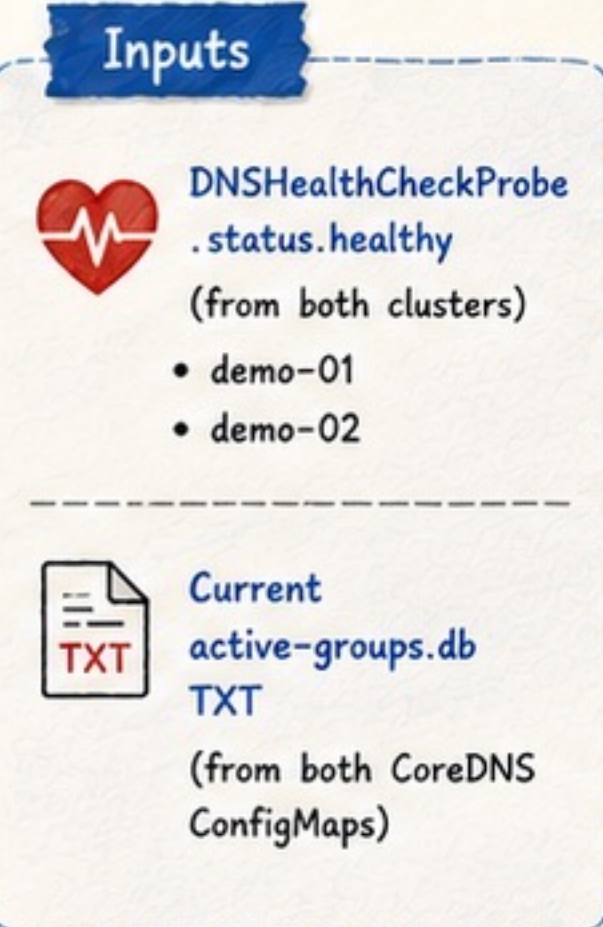


Key Takeaway

The controller is the **decision** and **coordination brain**, not a DNS data plane component.

Controller Flow

Continuously reconcile health signals to maintain the **active-groups TXT** control signal.



Controller Protections

-  Will not write empty groups (prevents total outage).
-  Will not patch when unchanged (avoids unnecessary reloads).
-  Health probes provide true application health, not affected by DNSRecord active group membership.

Output (Effect)

-  active-groups TXT updated in both CoreDNS sources
-  CoreDNS reloads zone file
-  DNS Operator recalculates DNSRecord
-  DNS answers change inside delegated DNS

Key Takeaway

The controller is a **decision and coordination brain**. It safely updates the **active-groups control signal** so that DNS answers always reflect the currently healthy clusters.

Internal zone (control):
 active-groups.txt.kuadrant-system.svc.cluster.local
 Used only for controller communication.

Legend: → Normal flow

---> Conditional / loop

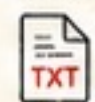
→ Decision



CoreDNS (NS targets)



DNSHealthCheckProbe



TXT active-groups TXT

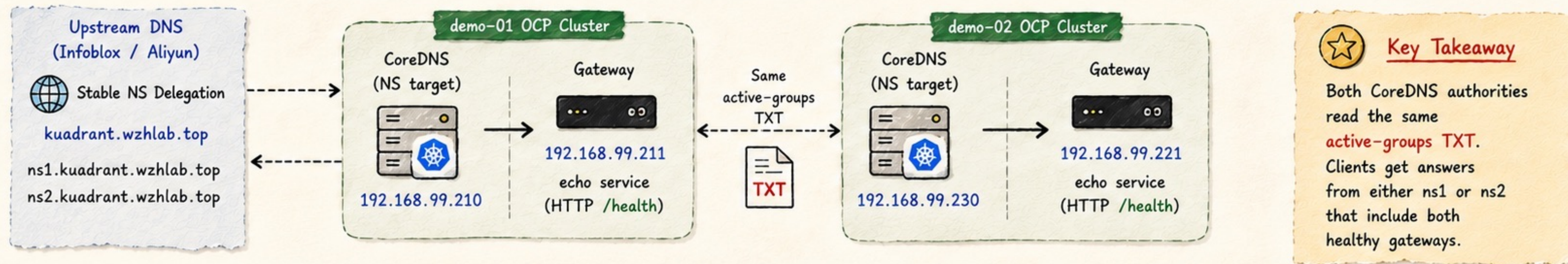
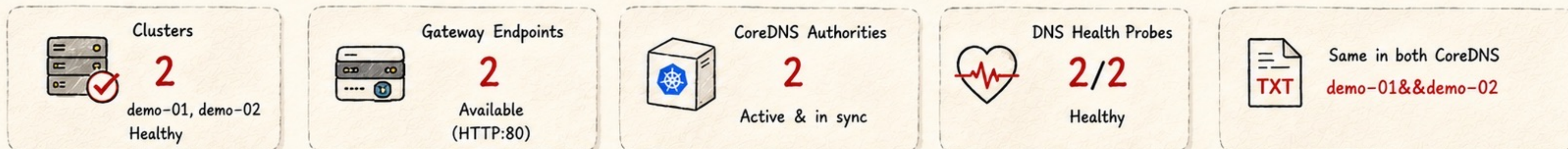
Baseline Validation (All Healthy)

Both clusters are healthy and active. All components operate as expected.



Baseline:

active groups: **demo-01&&demo-02**



1 Active-Groups TXT Evidence (Both CoreDNS)

@192.168.99.210 (ns1)

```
$ dig +tcp @192.168.99.210 \
kuadrant-active-groups.echo.kuadrant.wzhlab.top TXT +short
```

"version=1;groups=demo-01&&demo-02"

@192.168.99.230 (ns2)

```
$ dig +tcp @192.168.99.230 \
kuadrant-active-groups.echo.kuadrant.wzhlab.top TXT +short
```

"version=1;groups=demo-01&&demo-02"

✓ Both authoritative CoreDNS sources return the **same** active-groups TXT.

2 Application Access Evidence (Both Gateways)

Gateway 192.168.99.211 (demo-01)

```
$ curl -H "Host: echo.kuadrant.wzhlab.top" \
http://192.168.99.211/
```

demo-01 via Connectivity Link

Gateway 192.168.99.221 (demo-02)

```
$ curl -H "Host: echo.kuadrant.wzhlab.top" \
http://192.168.99.221/
```

demo-02 via Connectivity Link

✓ Both gateways respond and serve the application successfully.

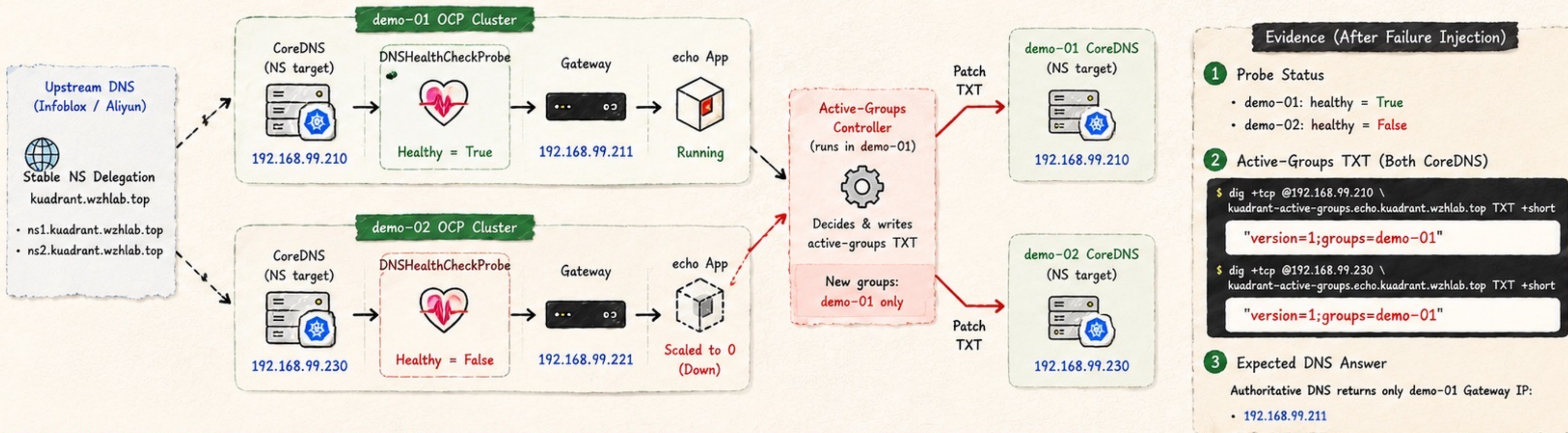
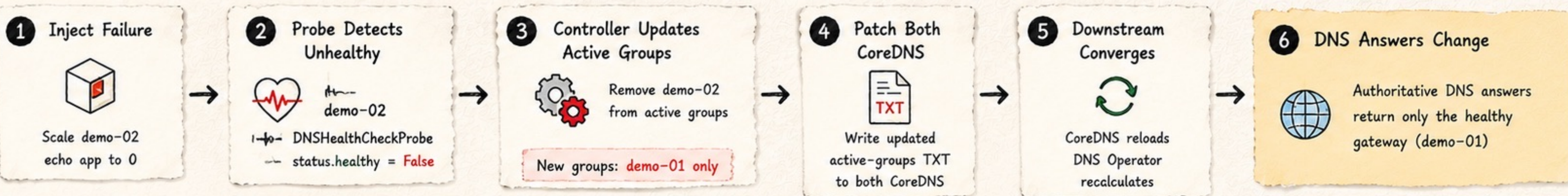


Baseline confirmed: 2 clusters healthy, active-groups = **demo-01&&demo-02**, both gateways serving traffic.

— Active Path ----> Delegation / Sync

Failure Injection

Failure starts at the **application layer**, not the DNS layer.



Key Takeaway:

Failure starts at the **application**. → Probe becomes **false** → Controller removes **demo-02** → Both CoreDNS get new **active-groups gateway**.

→ Normal signal flow

→ Control / update flow

-----> Unhealthy signal

Failover Result (demo-02 Unhealthy)

★ Only healthy gateway returned

Even with the **same upstream NS** delegation, both CoreDNS authorities return **only the healthy gateway**.



Active Clusters

1/2

demo-01 healthy
demo-02 unhealthy



Health Probes

1/2

demo-01: **True**
demo-02: **False**



active-groups TXT

1 Group

demo-01 only



Authoritative Sources

2

Both CoreDNS
in sync



Healthy Gateway

1

Serving all traffic

Current State Overview

demo-01



Healthy
Gateway 192.168.99.211

demo-02



Unhealthy
Gateway 192.168.99.221

Upstream DNS Delegation (Unchanged)



kuadrant.wzhlab.top
NS delegation stays the same

- ns1.kuadrant.wzhlab.top → 192.168.99.210
- ns2.kuadrant.wzhlab.top → 192.168.99.230

Stable NS delegation ✓

What Happened

- ✓ demo-02 echo app scaled to 0
- ✓ Demo-02 probe became **False**
- ✓ Controller removed demo-02 from active groups
- ✓ Both CoreDNS updated active-groups TXT

DNS Answer Result (Failover in Effect)

All queries return the healthy gateway from **both CoreDNS** authorities.

192.168.99.211

demo-01 Gateway (Healthy)

Evidence: DNS Queries (A Record for echo.kuadrant.wzhlab.top)

Query to ns1 CoreDNS (192.168.99.210)

```
$ dig +tcp @192.168.99.210 echo.kuadrant.wzhlab.top A +short
```

192.168.99.211

Query to ns2 CoreDNS (192.168.99.230)

```
$ dig +tcp @192.168.99.230 echo.kuadrant.wzhlab.top A +short
```

192.168.99.211



Why This Works

The active-groups TXT is the control signal for which clusters are active.
Both CoreDNS authorities read the same TXT and return only the healthy gateway IP(s).
DNS answer changes inside the delegated DNS, not in the parent DNS.

End-to-End Flow (After Failover)



Client / Recursive DNS

Upstream DNS (Infoblox / Aliyun)
NS delegation for kuadrant.wzhlab.top (unchanged)

ns1 CoreDNS
192.168.99.210

ns2 CoreDNS
192.168.99.230

Return: 192.168.99.211
(demo-01 Gateway only)

Gateway 192.168.99.211
echo service (/health) ✓ **Healthy**



Key Takeaway

NS delegation remains stable.
Connectivity Link updates active-groups TXT.
Both CoreDNS return **only the healthy gateway**.



Demo environment: demo-01 OCP, demo-02 OCP, Aliyun DNS simulating Infoblox delegation



Delegated subdomain: kuadrant.wzhlab.top

Application host: echo.kuadrant.wzhlab.top

Recovery Result (demo-02 Recovered)

After the demo-02 application **recovered**, it is automatically **added back** to active groups.



Restored

active groups: **demo-01&&demo-02**



Active Clusters

2/2

demo-01, demo-02
Healthy



Health Probes

2/2

demo-01: True
demo-02: True



active-groups TXT

2 Groups

demo-01&&demo-02



Authoritative Sources

2

Both CoreDNS
in sync



DNSRecord
Active

2

Both Active=True



Healthy Gateways

2

Both gateways
serving traffic

Recovery Timeline

- 1 demo-02 application scaled back up
- 2 DNSHealthCheckProbe becomes healthy=true
- 3 Controller adds demo-02 back to active groups
- 4 active-groups TXT returns to "demo-01&&demo-02"
- 5 DNSRecord Active=True in both clusters

Evidence: active-groups TXT (Both CoreDNS)

Query to ns1 CoreDNS (192.168.99.210)

```
$ dig +tcp @192.168.99.210  
kuadrant-active-groups.echo.kuadrant.wzhlab.top TXT +short
```

"version=1;groups=demo-01&&demo-02"

Query to ns2 CoreDNS (192.168.99.230)

```
$ dig +tcp @192.168.99.230  
kuadrant-active-groups.echo.kuadrant.wzhlab.top TXT +short
```

"version=1;groups=demo-01&&demo-02"

Evidence: application access (Both Gateways)

Gateway 192.168.99.211 (demo-01)

```
$ curl -H "Host: echo.kuadrant.wzhlab.top" \  
http://192.168.99.211/
```

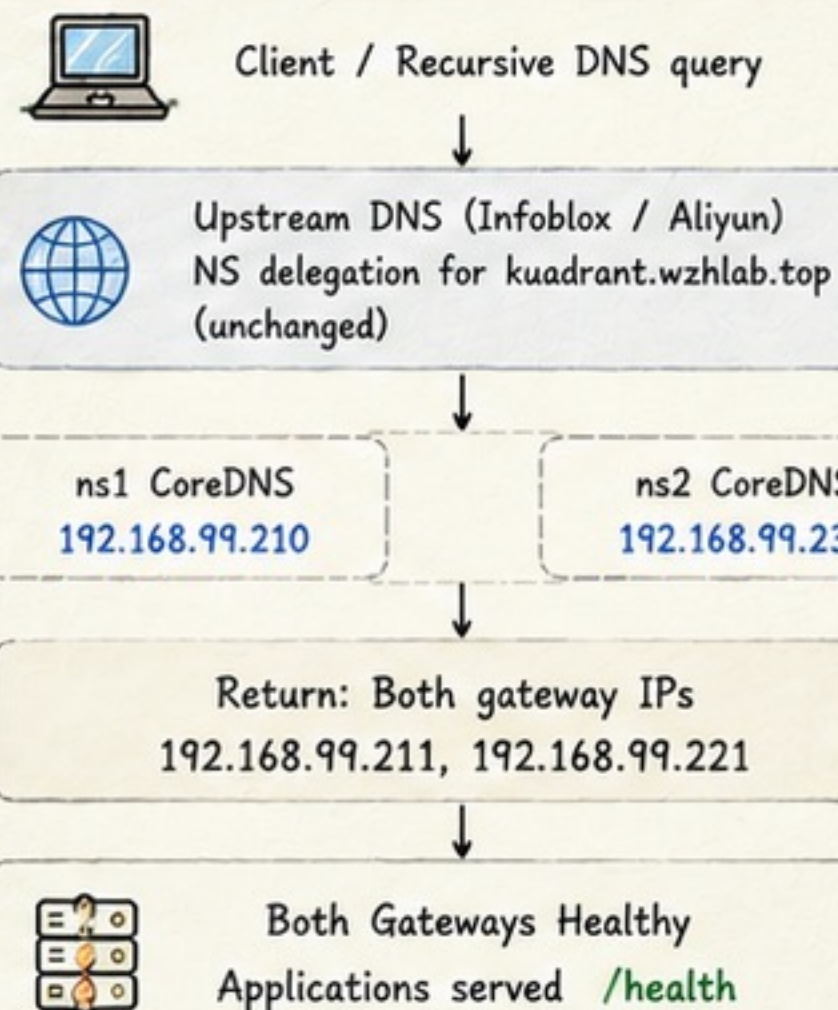
demo-01 via Connectivity Link

Gateway 192.168.99.221 (demo-02)

```
$ curl -H "Host: echo.kuadrant.wzhlab.top" \  
http://192.168.99.221/
```

demo-02 via Connectivity Link

End-to-End Flow (After Recovery)



Key Takeaway

When the application recovers, probe becomes healthy → controller adds demo-02 back → TXT returns to demo-01&&demo-02 → both CoreDNS return both gateway IPs again.

Legend

--- Delegation / Sync

— Normal flow

→ Controller action

End-to-End Failover Sequence (demo-02 Failure)

From application failure to DNS answer change – **what happens and what does NOT change.**

Key Principles



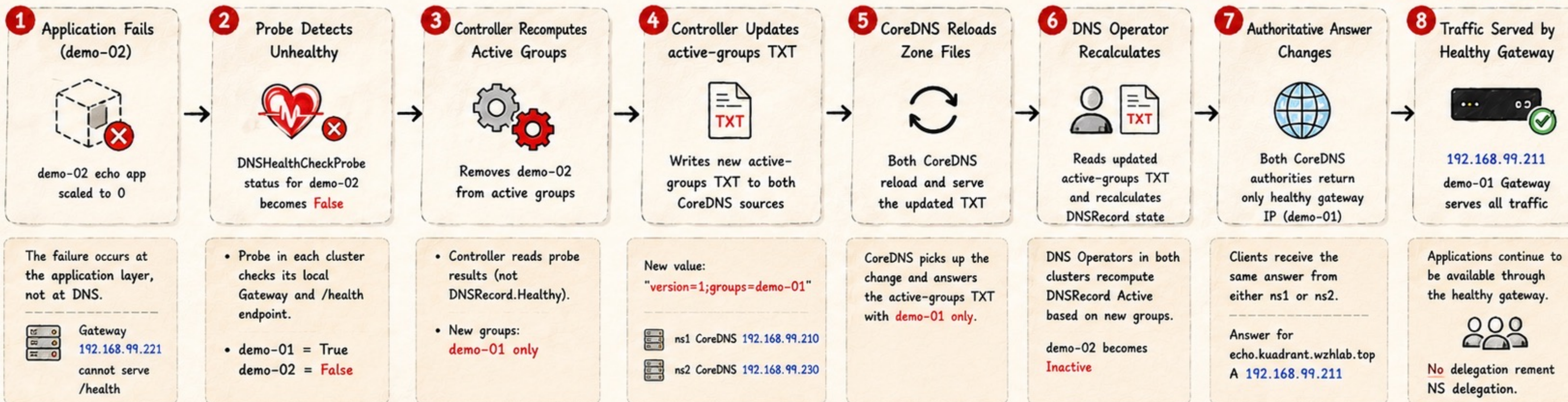
NS records do not switch.



Controller does not write application A records directly.



DNS Operator recalculates DNSRecord state from **active-groups**.



Upstream (Unchanged)



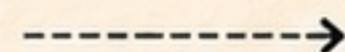
Infoblox / Aliyun (Parent DNS)

Stable NS delegation kuadrant.wzhlab.top

- ns1.kuadrant.wzhlab.top
- ns2.kuadrant.wzhlab.top



Stable NS Delegation (Does Not Change)



ns1 CoreDNS (demo-01) 192.168.99.210

ns2 CoreDNS (demo-02) 192.168.99.230

Application Gateways

demo-01 (Healthy)



192.168.99.211
Active

demo-02 (Unhealthy)



192.168.99.221
Inactive



Outcome

Failover completed.
Only healthy gateway is returned by DNS.

NS delegation remains stable.
DNS answer changes **inside** the delegated DNS.

Legend: → Normal flow → Controller / update flow ----> Delegation (unchanged)



Bottom line: Application failure → Probe false → Controller updates active-groups → CoreDNS reloads → DNS Operator recalculates → DNS answer changes.

ACM Positioning

ACM adds governance. It should **NOT** be in the DNS query path.

PHASE 1 (Now)

Direct failover path validated end-to-end

VALIDATED
NOW



NS points to managed-cluster CoreDNS

ns1: 192.168.99.210 (demo-01 CoreDNS)

ns2: 192.168.99.230 (demo-02 CoreDNS)



Controller runs in OCP PoC form

Reads probes and writes active-groups TXT to both CoreDNS sources



Direct failover path is validated

Only healthy gateway IP is returned; recovery is automatic.



Fewer moving parts

Simple, observable, and easier to operate.



Goal of Phase 1: Prove the solution works end-to-end with minimal dependencies and clear control.

VS

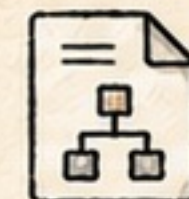
PHASE 2 (Future Enhancement)

ACM governs distribution and compliance

FUTURE
ENHANCEMENT



ACM distributes and governs config
ACM (Hub) distributes CoreDNS ConfigMap and controller manifests to clusters.



ACM manages Placement and ManifestWork
Placement chooses clusters; ManifestWork delivers manifests consistently.



Hub provides audit, policy, global visibility
Central audit, policy enforcement, and multi-cluster visibility.



Better operational governance
Standardized change management, compliance, and reporting.

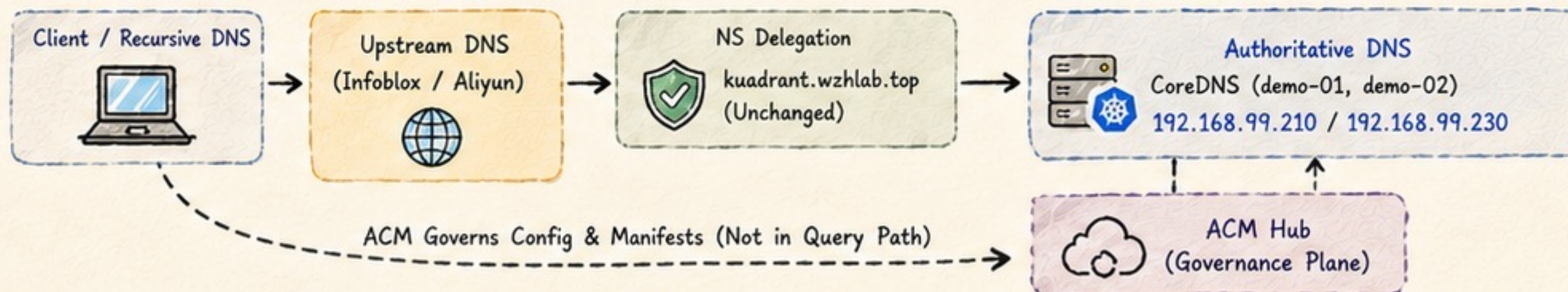


Goal of Phase 2: Scale with governance and consistency, without changing the DNS failover design.



Key Principle: DNS Query Path Remains Unchanged

Even with ACM, NS delegation still points to CoreDNS (or Infoblox / Aliyun), **NOT** to the ACM Hub.



Production Risks

PoC proves the path works.
Production requires strengthening multiple areas.

#	Risk	Current State (PoC / Demo)	Production Recommendation
1	 Controller HA	 Single replica in demo-01	- Deploy controller with leader election - Run multiple replicas across nodes - Use PodDisruptionBudget and proper probes
2	 Remote kubeconfig	 PoC token + insecure TLS	- Use correct CA validation - Short-lived token via ServiceAccount - Least-privilege RBAC (read-only) - Automatic token rotation
3	 ConfigMap projection delay	 Tens of seconds observed	- Sidecar-managed shared zone file (atomic write + notify CoreDNS) - Or use production DNS provider (Infoblox / Aliyun)
4	 Health decision	 Simple probe state	- Debounce probe flaps - Hysteresis (N consecutive successes/failures) - Minimum healthy cluster policy (e.g., ≥ 1) - Optional quorum / time-based rules
5	 Audit & observability	 Controller stdout	- Emit Kubernetes Events - Expose metrics (Prometheus) - Structured audit logs (who/what/when) - Alerting on state changes and errors
6	 CoreDNS reachability	 Lab private IPs (e.g., 192.168.99.*)	- Use enterprise-reachable Load Balancer or Anycast - Or provide secure forwarding path from recursive DNS

Demo Environment Snapshot

demo-01 OCP Cluster

 CoreDNS (ns1)
192.168.99.210

 Gateway
192.168.99.211

 App (echo)
192.168.99.211

demo-02 OCP Cluster

 CoreDNS (ns2)
192.168.99.230

 Gateway
192.168.99.221

 App (echo)
192.168.99.221

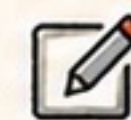
 **Delegated Domain**
kuadrant.wzhlab.top
Stable NS delegation
(unchanged)

Upstream DNS
Infoblox / Aliyun

Common Pitfalls to Avoid



Letting ACM Hub
in the DNS
query path



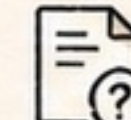
Controller writing
application A records
directly



Ignoring propagation
delay and causing
flapping answers



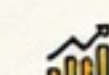
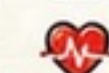
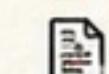
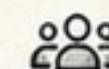
Over-privileged
ServiceAccounts
in clusters



No audit trail
for DNS answer
changes

Production Readiness Checklist

- ☒ Controller HA with leader election and PDB
- ☒ Secure remote access: CA, short-lived token, least-privilege RBAC, rotation
- ☒ Deterministic config delivery: sidecar or production DNS provider
- ☒ Robust health decision: debounce, hysteresis, min healthy policy
- ☒ Observability & audit: events, metrics, Prometheus, audit logs, alerting
- ☒ Enterprise DNS reachability: LB/Anycast or forwarding path



Key Takeaway: The architecture is valid. Production success depends on **HA**, **security**, **latency control**, **observability**, and **operational boundaries**.

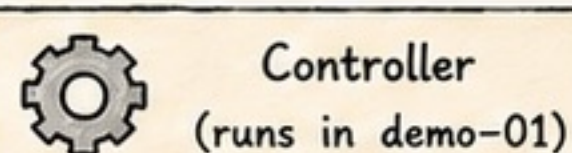
- PoC / Current State
- Next Step / Recommendation

Future Infoblox Integration

Same health source, different DNS write target.

1 Current (PoC)

Controller writes active-groups TXT to OCP CoreDNS



Write target

CoreDNS (ns1)
192.168.99.210



CoreDNS (ns2)
192.168.99.230



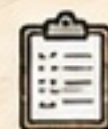
active-groups TXT
version=1;groups=demo-01&&demo-02
(updated by controller)



Delegated Zone
kuadrant.wzhlab.top
(NS to CoreDNS ns1/ns2)



Replace CoreDNS TXT write with Infoblox DTC / WAPI write

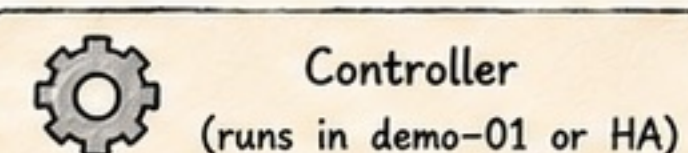


Evolution Checklist

- 1 Add Infoblox WAPI client.
- 2 Store Infoblox credential in least-privilege Secret.
- 3 Prefer DTC pool and LBDN member state updates.
- 4 Add ownership markers to avoid changing manual records.
- 5 Add dry-run, diff, rollback, audit, and approval controls.
- 6 Keep RHCL probes and Gateway state as the source of application health.

2 Future (Production Goal)

Controller writes to Infoblox DTC or authoritative DNS via WAPI



Write target

Infoblox DTC Pool



demo-pool

LBDN Members

- 192.168.99.211 (demo-01)
- 192.168.99.221 (demo-02)



WAPI

Update pool / member state via WAPI
(preferred over editing individual A records)



Delegated Zone (unchanged)
kuadrant.wzhlab.top
(NS remains stable)



Common Health Source
(Unchanged)



Applications
demo-01 / demo-02



DNSHealthCheckProbe
(per cluster)



Health Decision Logic
(controller)



Active Groups
(computed)



DNS Record State
(via DNS Operator)



DNS Answer Changes
(inside delegated DNS)



Goal: Move from CoreDNS TXT control signal to Infoblox authoritative DNS control, with stronger governance, safety, and enterprise-grade operations.



Principle: Stable NS delegation.
Health-driven answers. Safer writes.

Final Recommendation and Roadmap

From validated PoC to production-ready operations.



Three Key Lines (Keep Them Clear)

